

Chapter 7

Wireless and Mobile Networks

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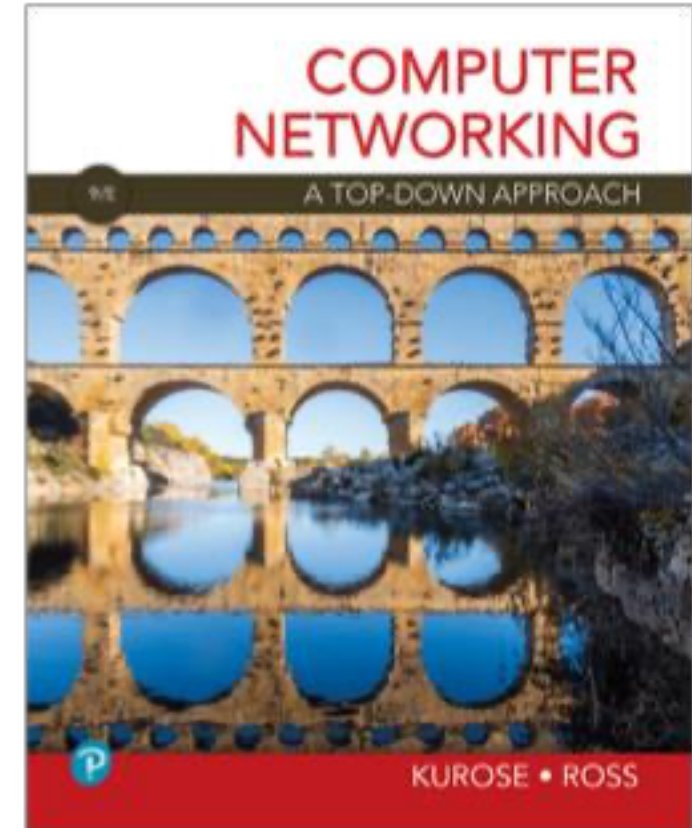
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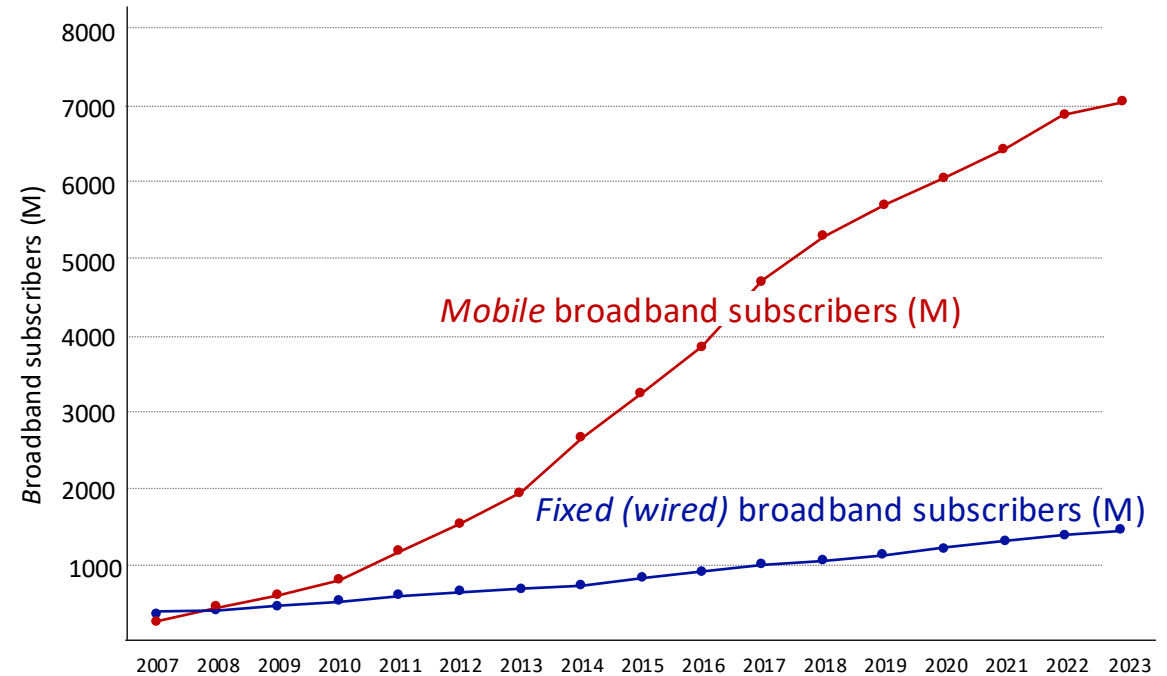
*Computer Networking: A
Top-Down Approach*

9th edition

Jim Kurose, Keith Ross
Pearson, 2025

Wireless and Mobile Networks: context

- more mobile-broadband-connected (cellular) devices than fixed-broadband-connected devices (5-1 in 2019)!
- wireless connectivity even great when WiFi users considered (80% of broadband homes use WiFi)
- 60% of Internet traffic from major web sites destined to mobile device

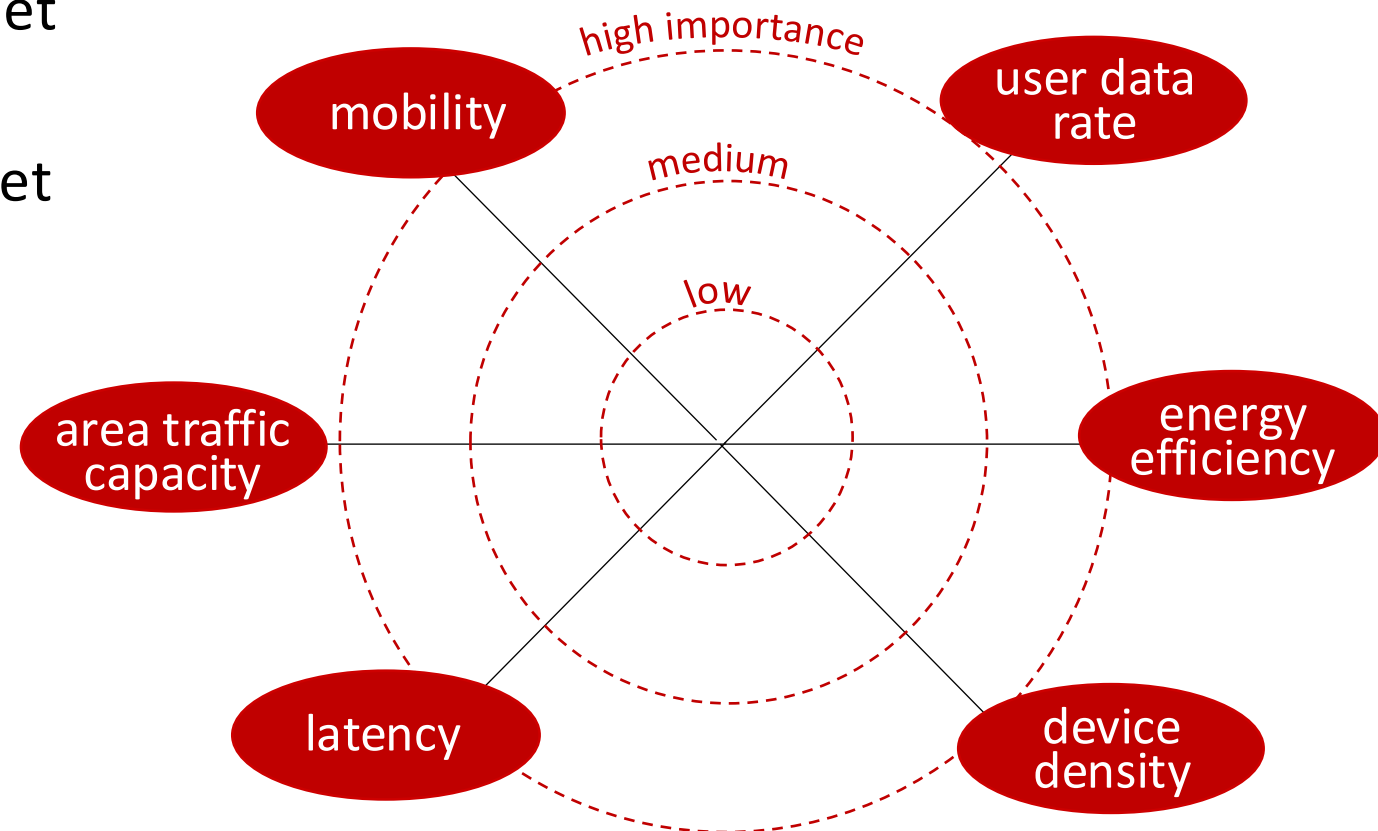


Wireless applications their needs

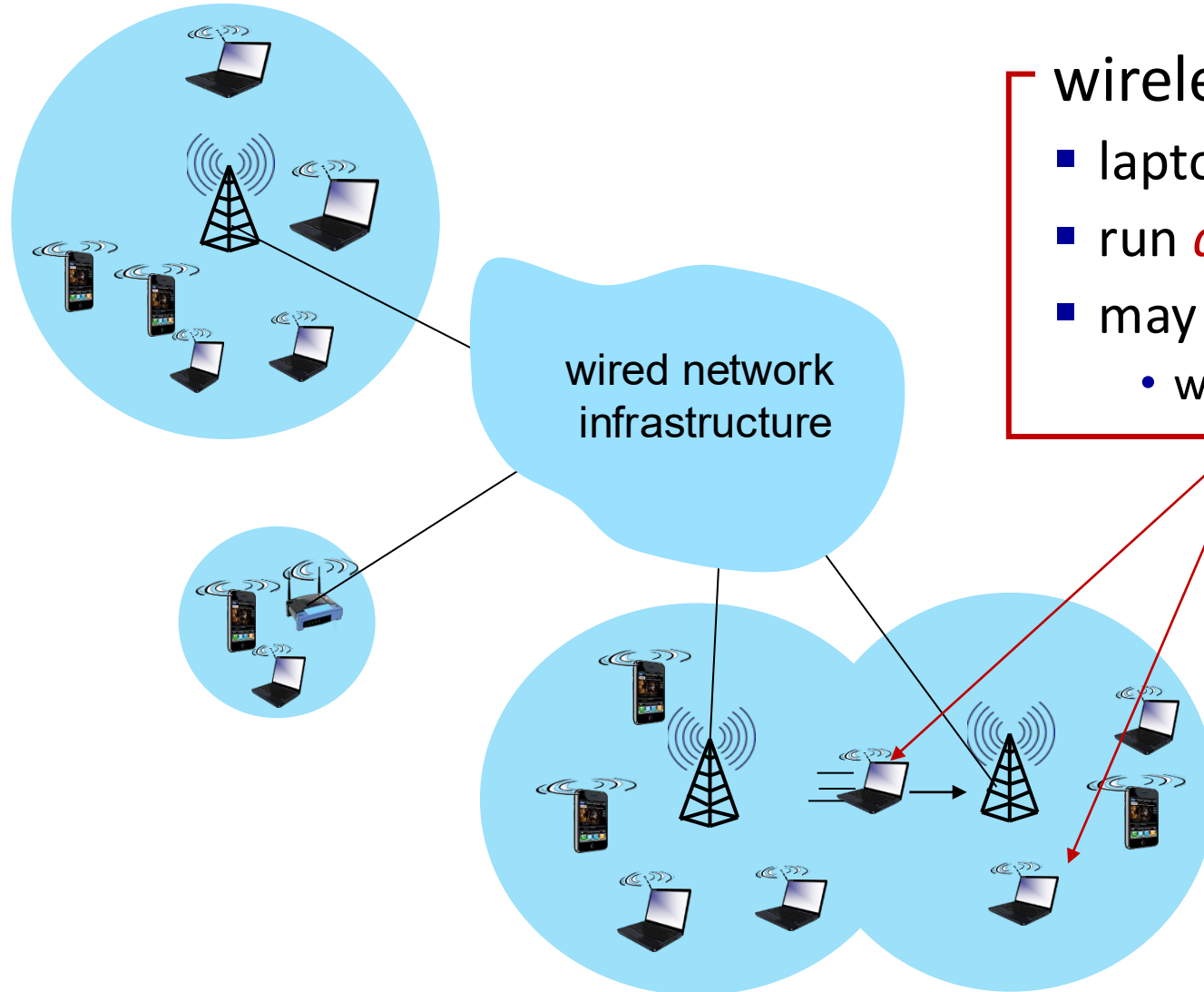
Six application areas :

- Wide-area Mobile Wireless Internet Access
- Local-area Mobile Wireless Internet Access
- Fixed Wireless Internet Access
- Satellite Networks for Internet Access and Sensing
- Cable replacement
- Internet of Things (IoT)

Application needs:

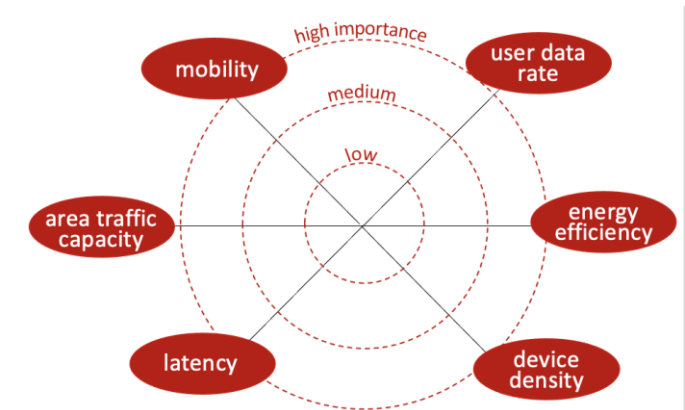


Elements of a wireless network

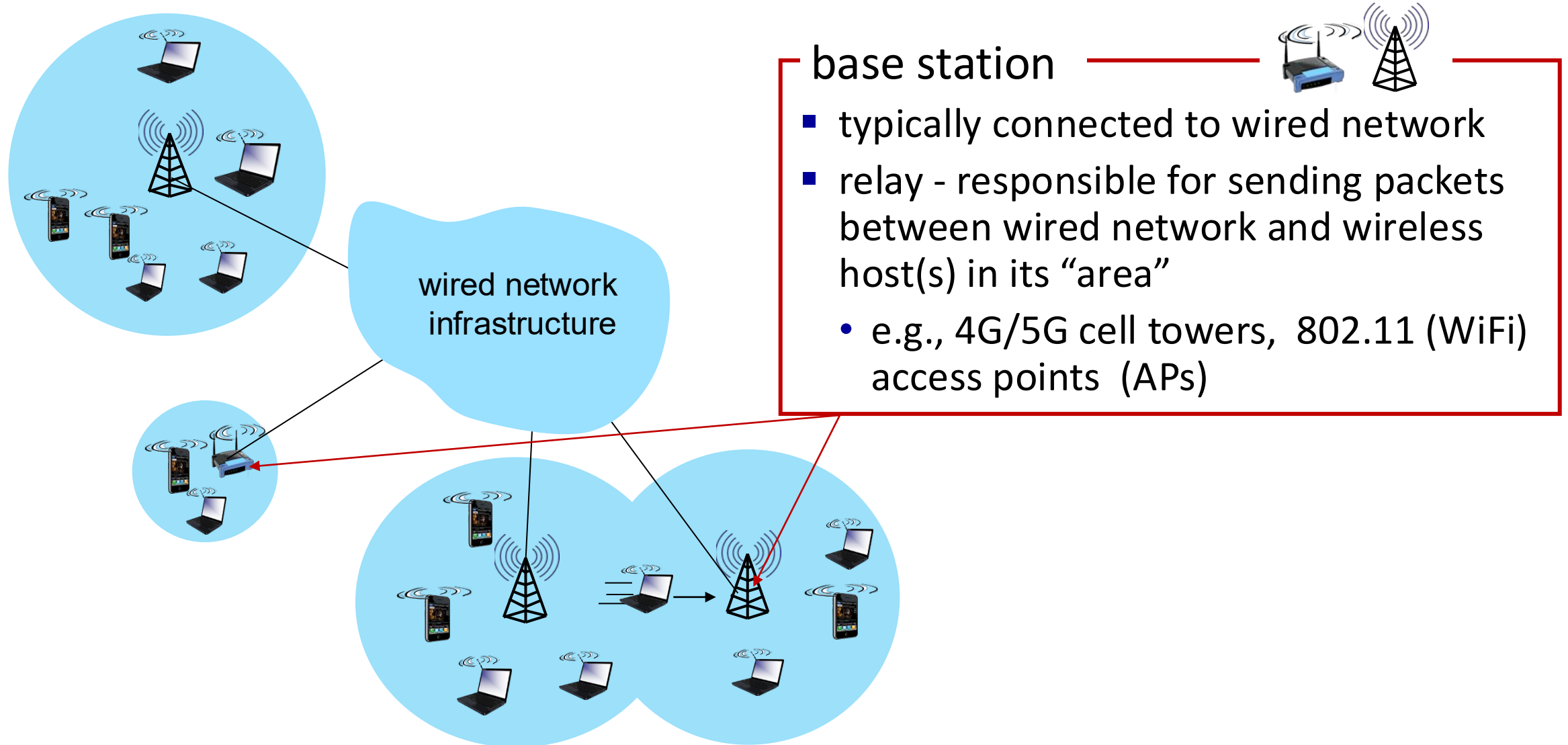


wireless hosts

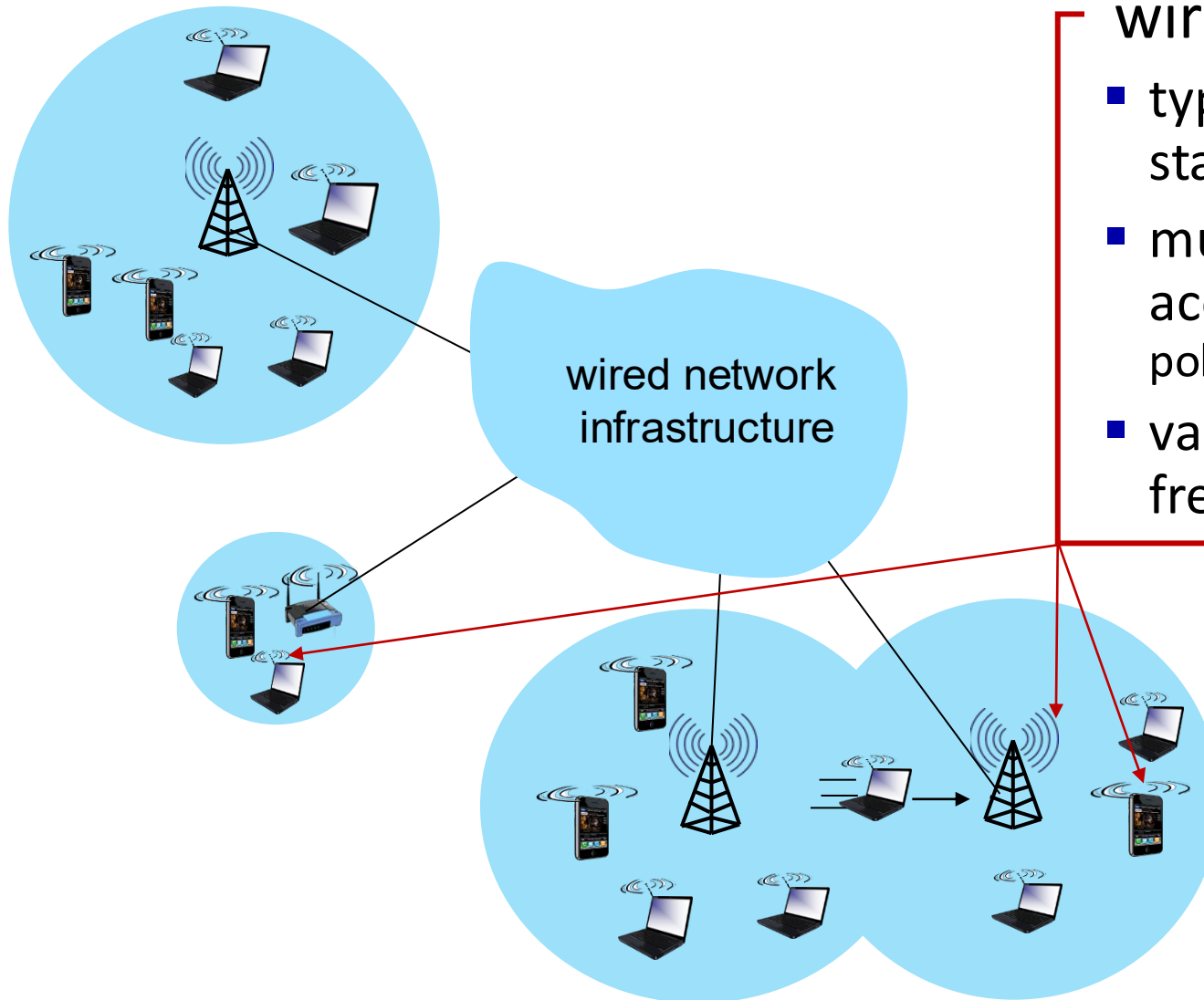
- laptop, smartphone, IoT
- run *applications*
- may be stationary (non-mobile) or mobile
 - wireless does *not* always mean mobility!



Elements of a wireless network



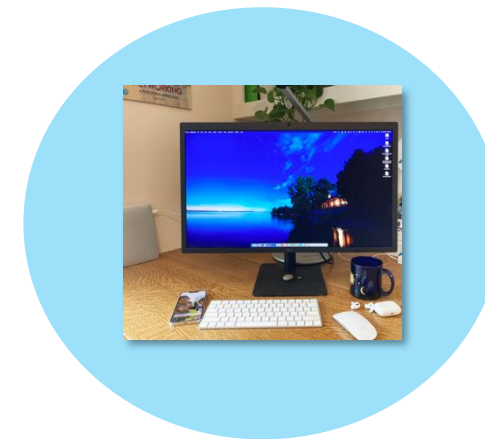
Elements of a wireless network



wireless link

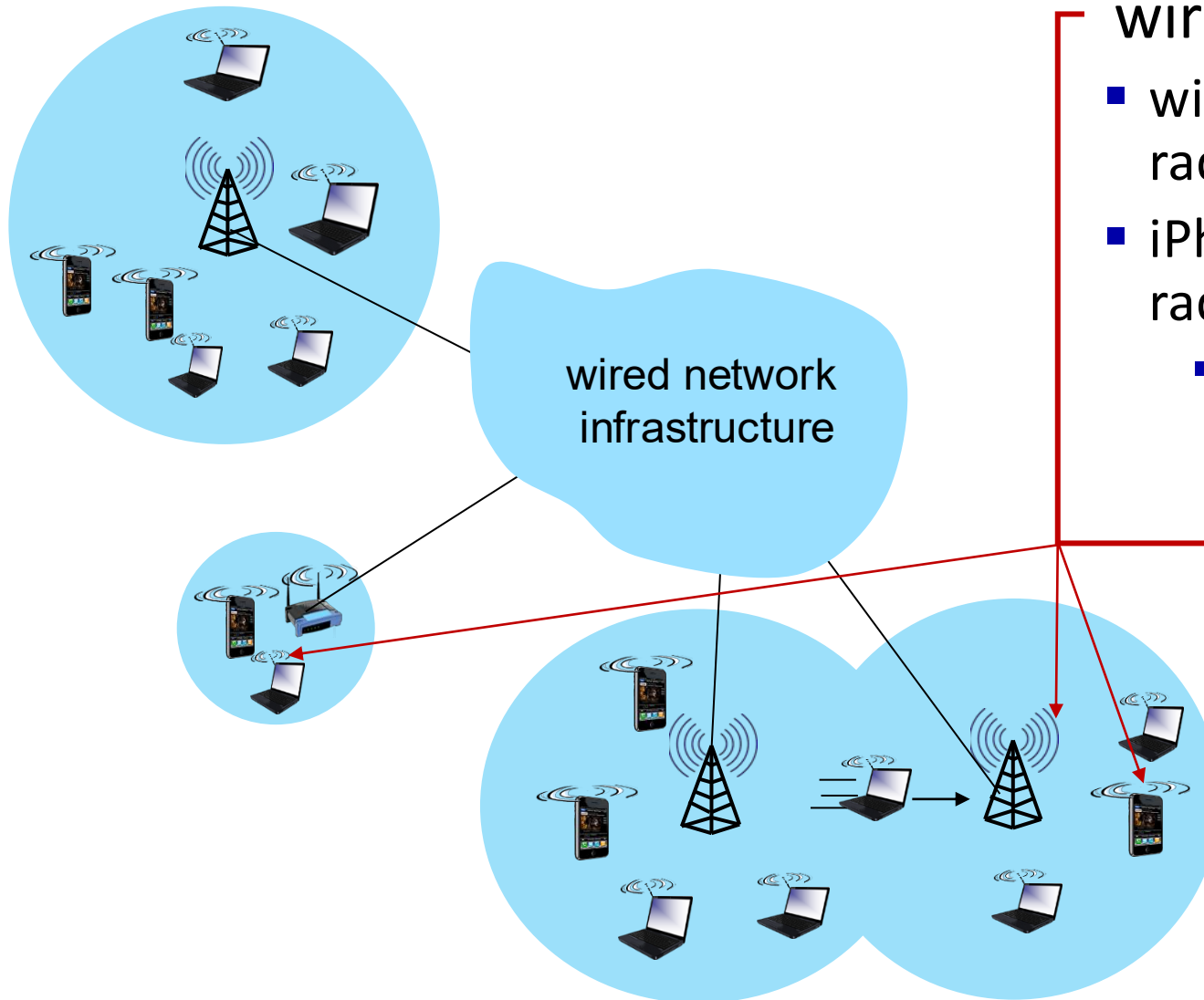


- typically used to connect device(s) to base station, also used as backbone link
- multiple access protocol coordinates link access (random access, FDMA, TDMA, CDMA, polling)
- various transmission rates and distances, frequency bands



*Ad hoc networks:
not all wireless
networks are
connected into a
larger network*

Elements of a wireless network



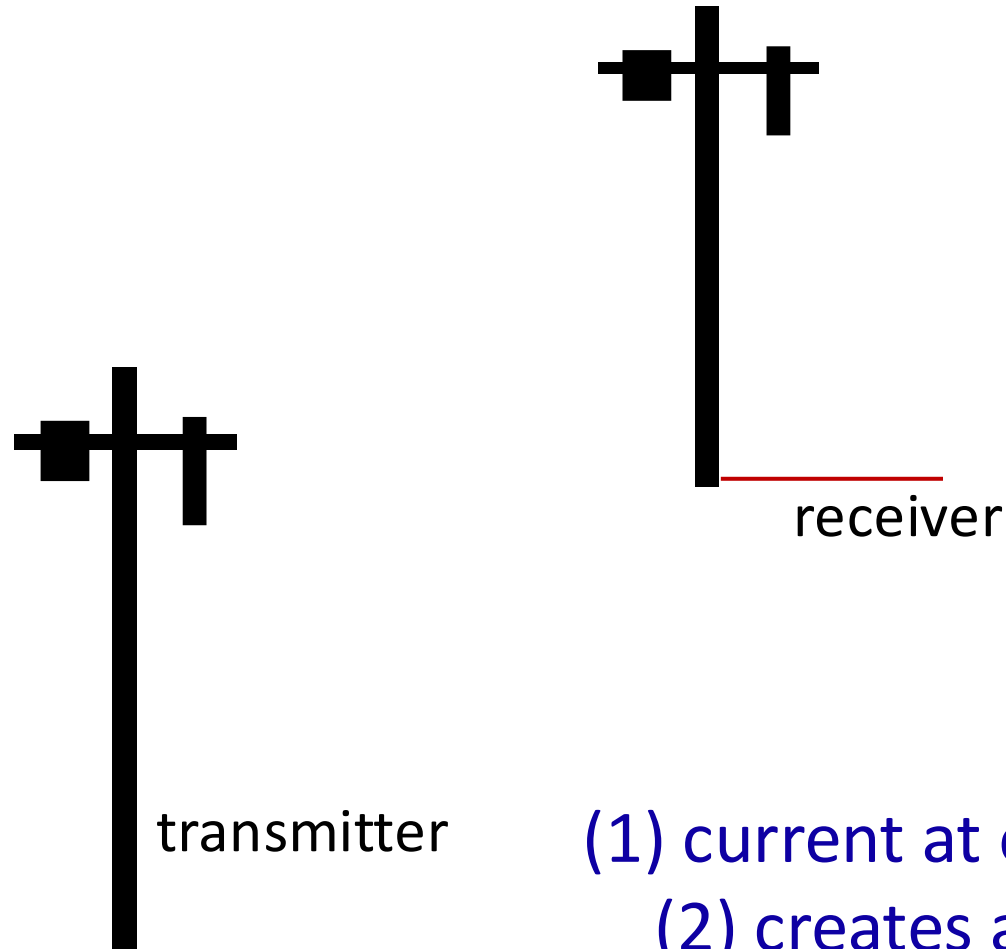
wireless device radio

- wireless device has different radios for different networks
- iPhone16: has ~11 different radios, many antennae
 - 5 different cellular radios, WiFi, Bluetooth, UWB, satellite NFC, GPS



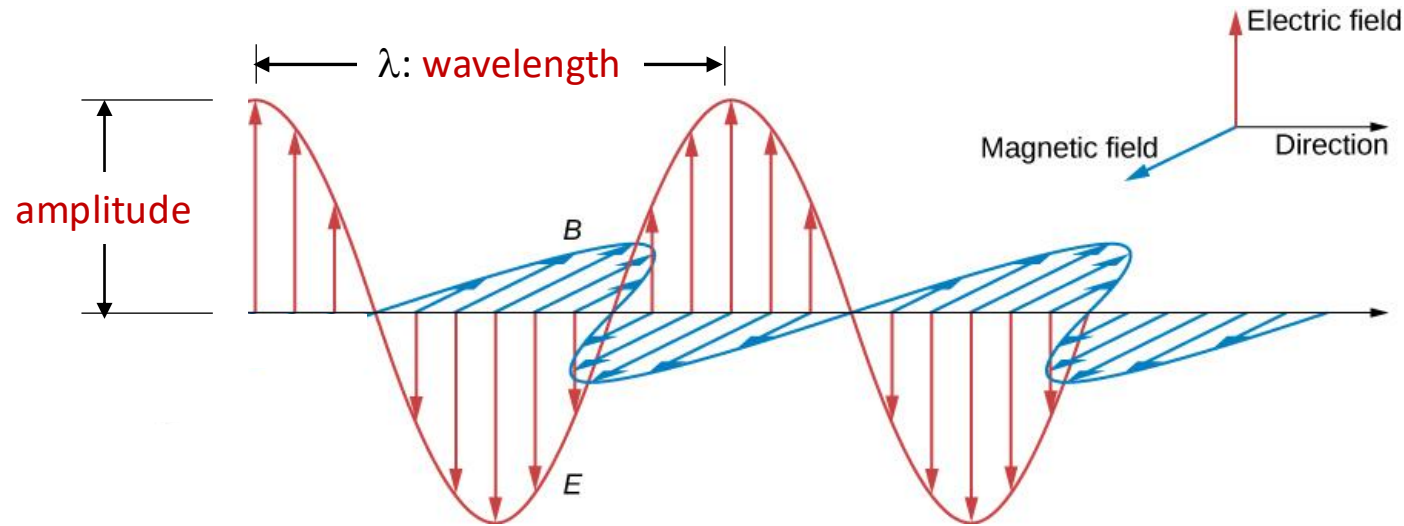
*Ad hoc networks:
not all wireless
networks are
connected into a
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Electromagnetics: enabling communication



- (1) current at one location (transmitter)
- (2) creates an EM field in space
- (3) induces a current at remote location (receiver)....

Radio basics: electromagnetic waves



wave propagation:

Electromagnetic wave has:

- **wavelength** (λ)
- **frequency**: speed of light / wavelength = c/λ
- **directionality** (of propagation)
- time-varying **amplitude**

Antennas

Antennas both transmit and receive radio waves

single antenna: old school

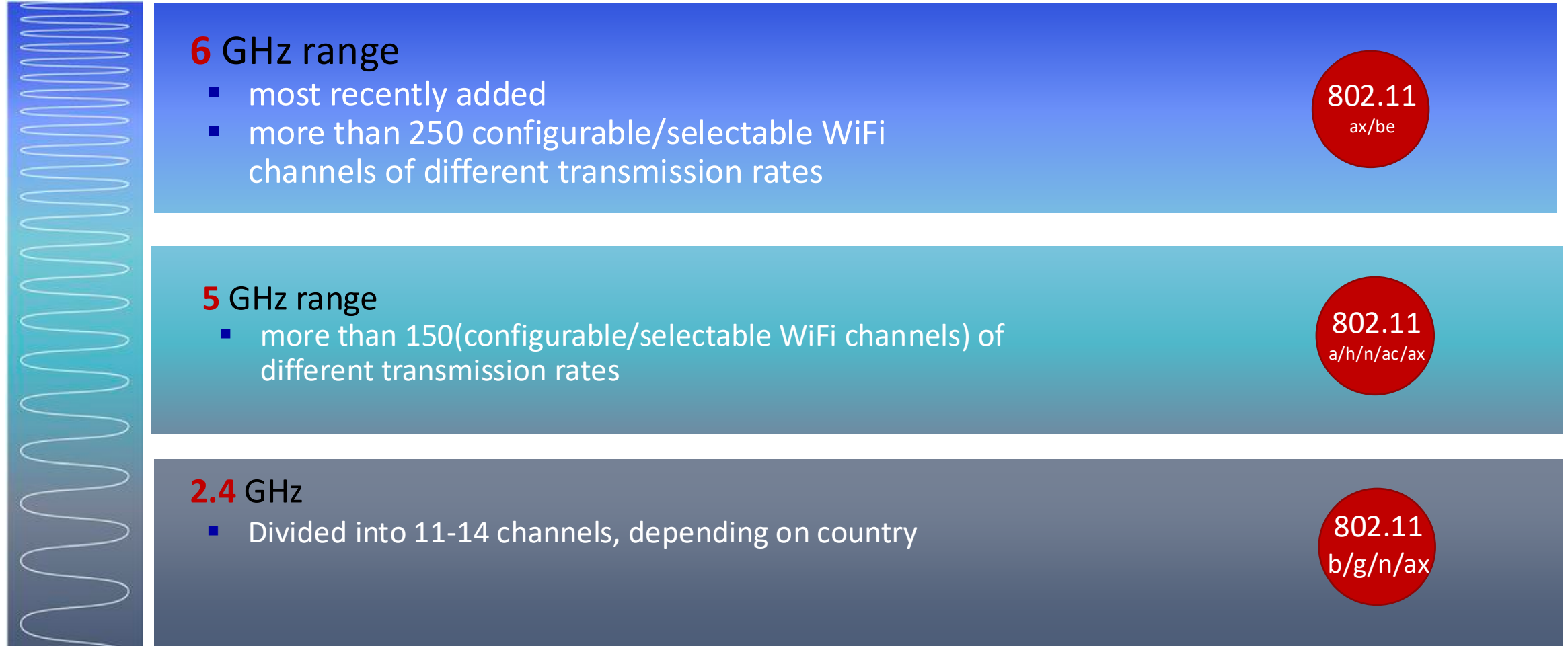
multiple antenna: common now in wireless networks

MIMO: Multiple input
multiple output
antenna



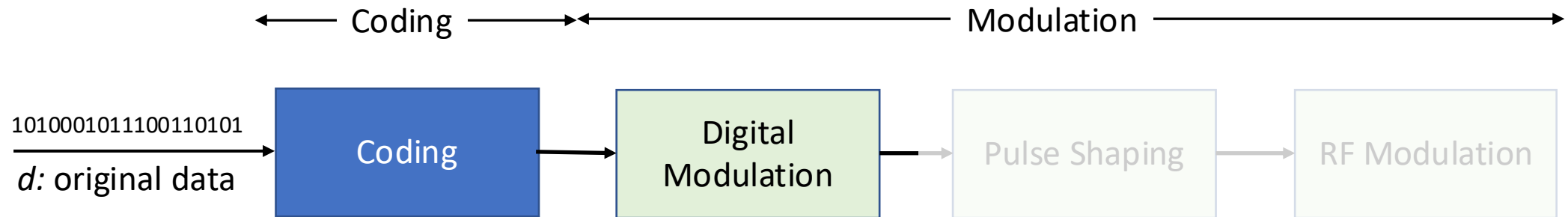
- Gigabit LTE with 4x4 MIMO
- Wi-Fi 6 (802.11ax) with 2x2 MIMO

WiFi spectrum bands



- other 802.11 WiFi spectrum bands, but not in widespread use

Physical Layer: coding and modulation



Add error detection, correction bits at physical layer

Choose digital modulation technique that maximizes number of reliably transmitted bits/sec based on observed channel impairments e.g., in family of Quadrature amplitude modulation (QAM)

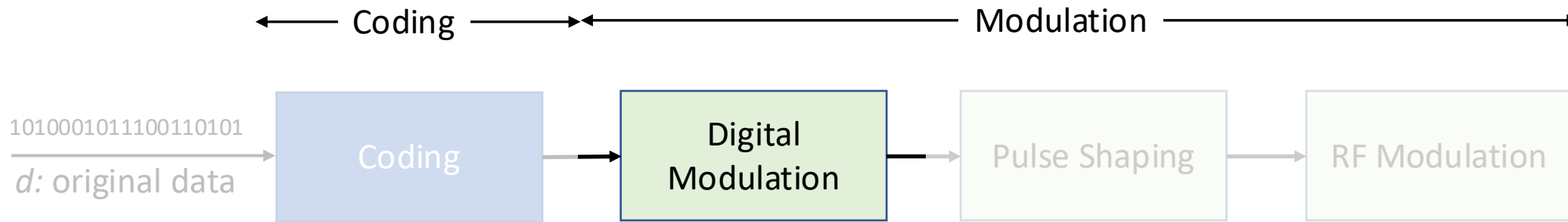
We won't cover this (look at ECE for courses)

match transmit signal bandwidth to channel bandwidth using pulse shaping

transmit signal as an electromagnetic wave over assigned carrier frequency

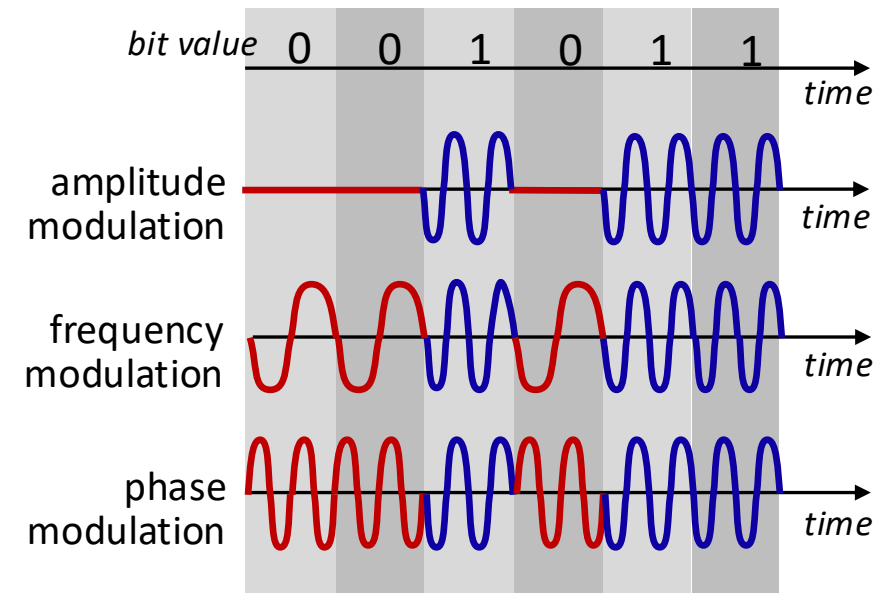
Carrier: waveform at a given frequency, f_c (carrier frequency), modified (modulated) with an information-bearing signal

Modulation

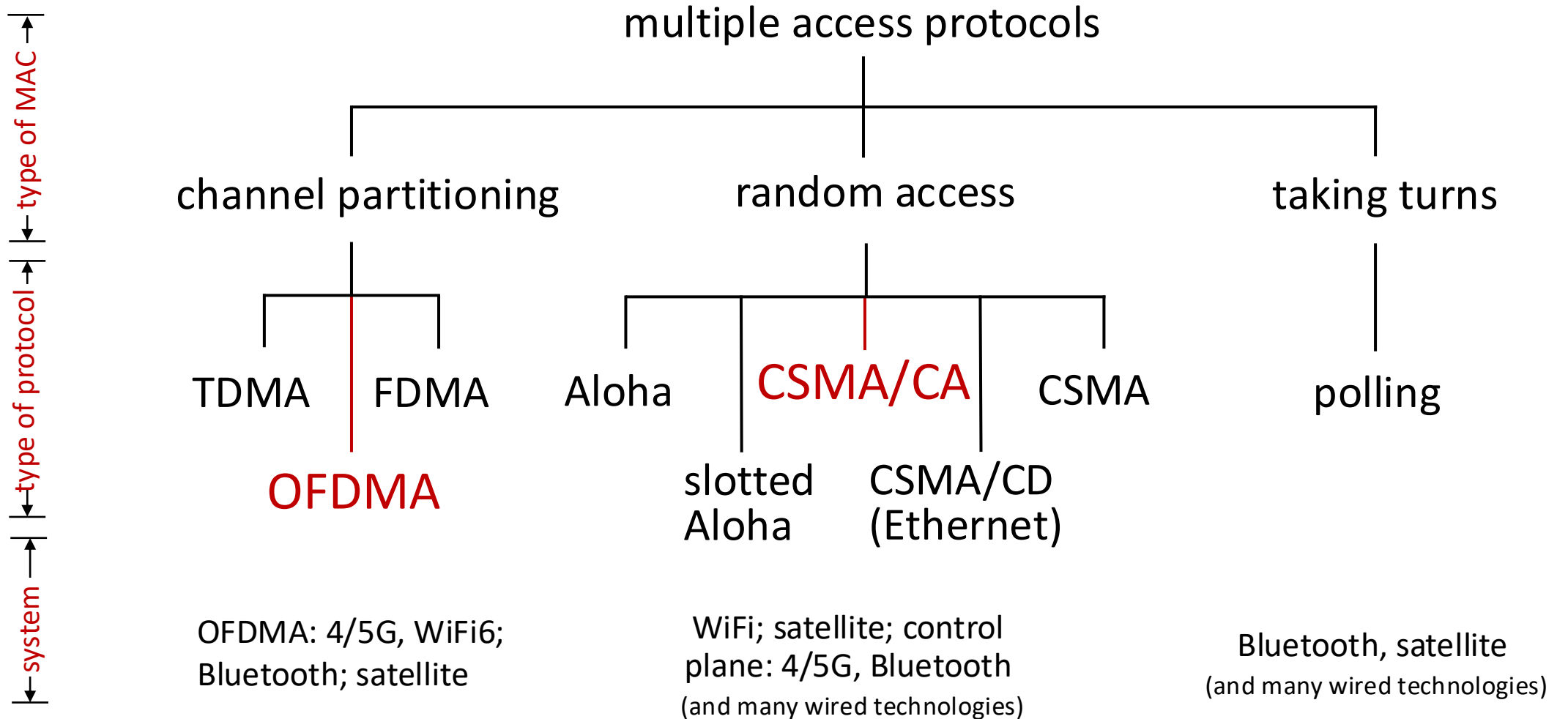


digital modulation

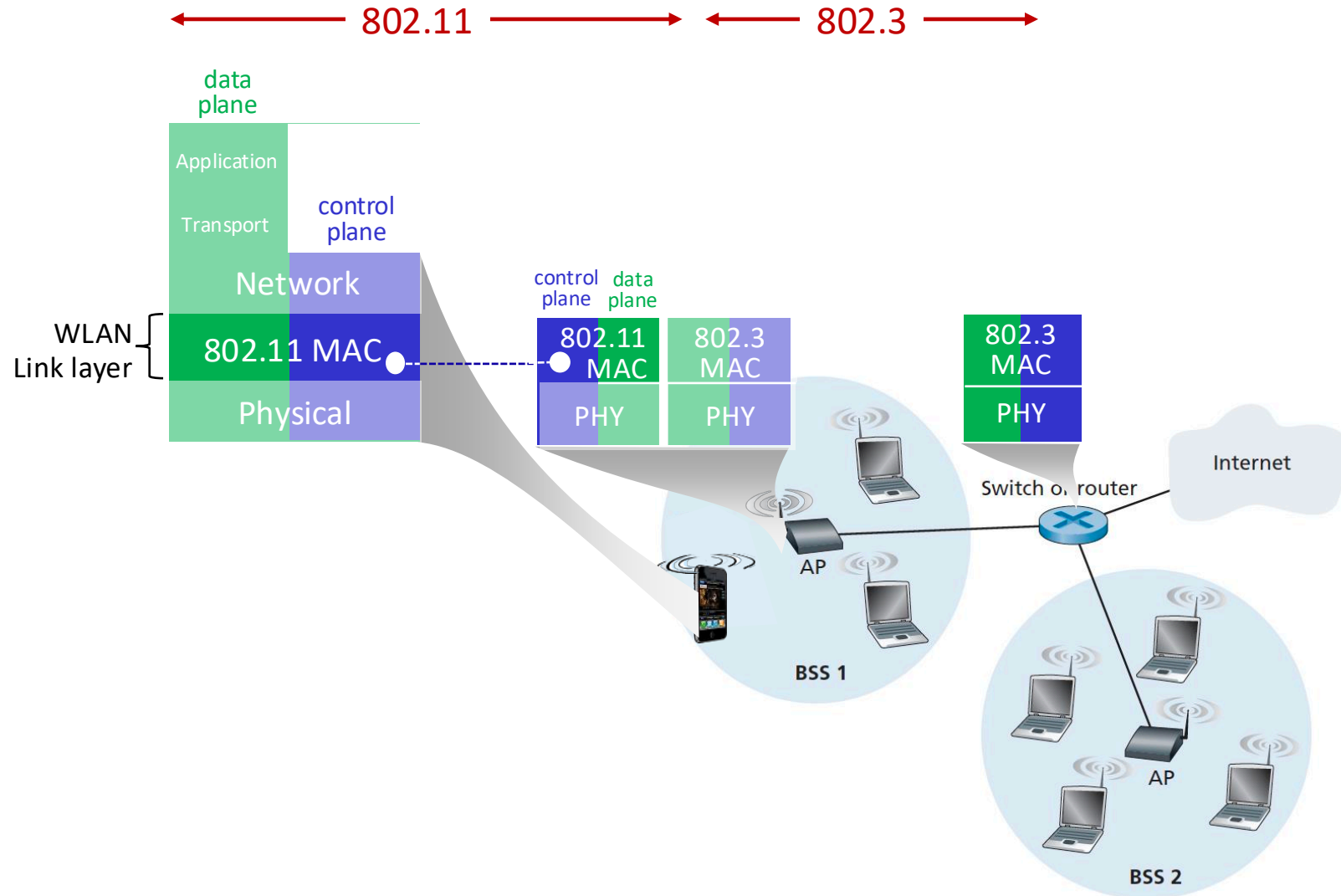
encoding bits (1's, 0's) into analog RF carrier signal by changing (“**modulating**”) a characteristic of the RF carrier signal: *amplitude, frequency, or phase*.



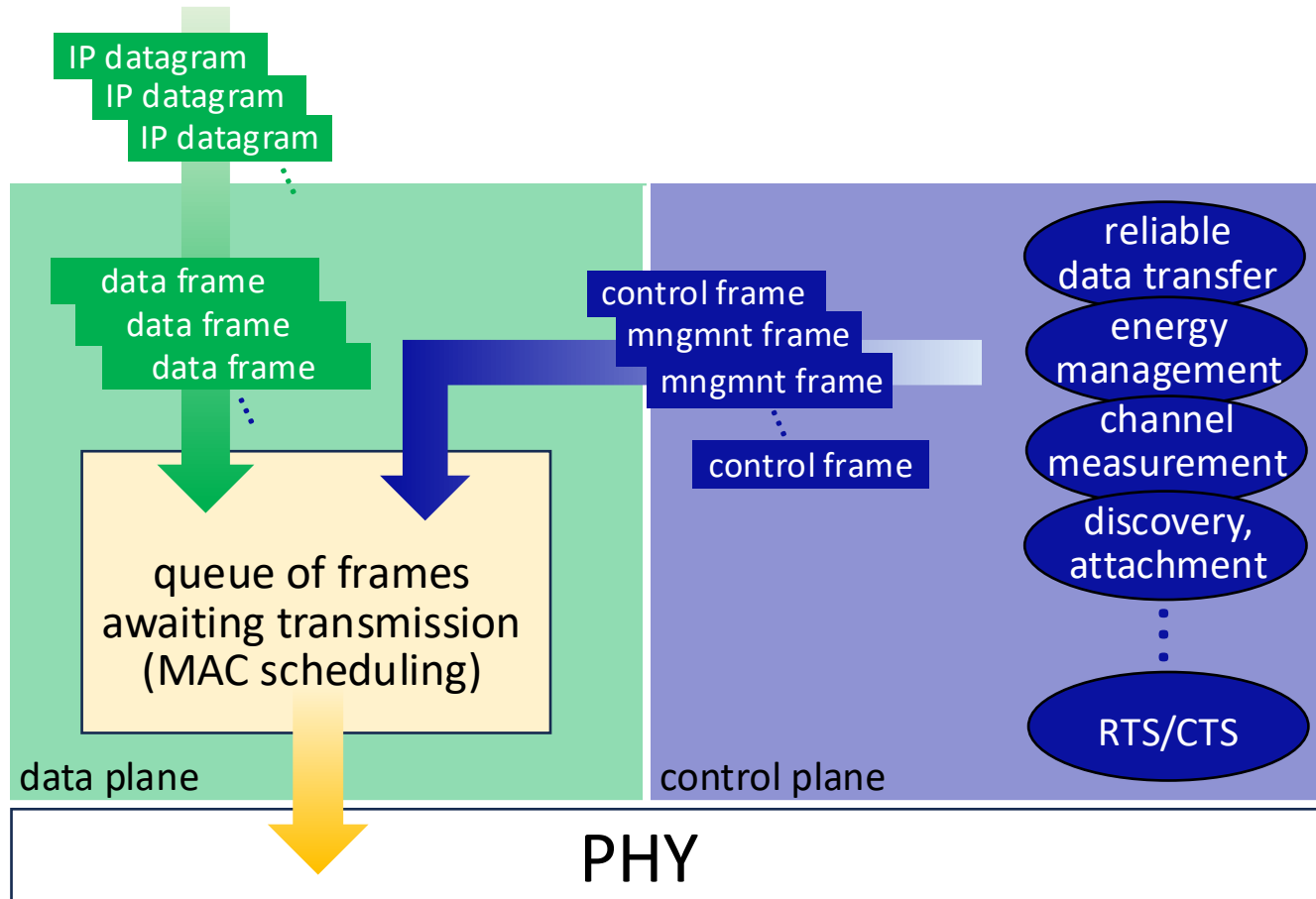
Recall: MAC protocols



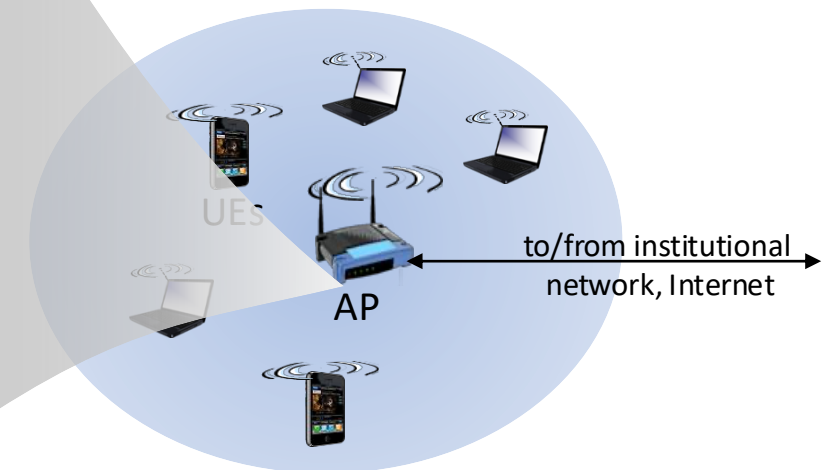
802.11 wireless link layer



802.11 downlink (AP-to-device) frame flow

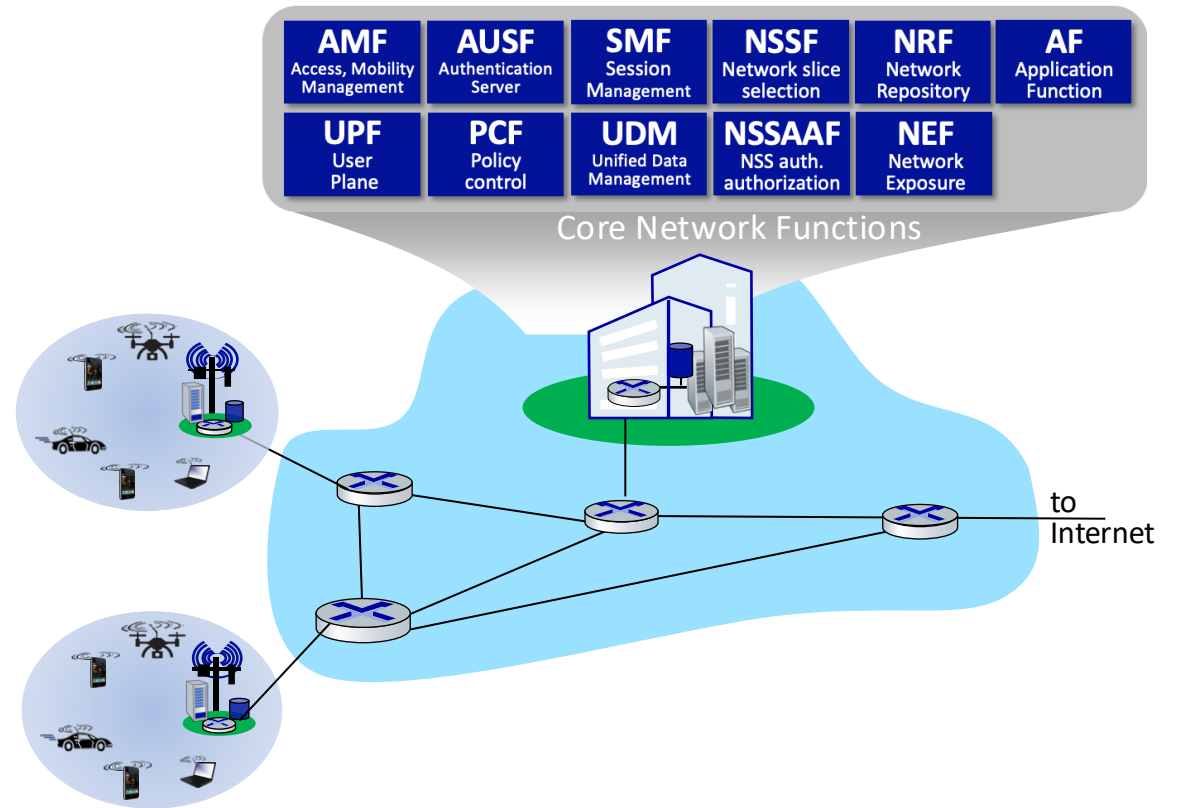


- data frames: containing IP datagrams
- Control, management frames



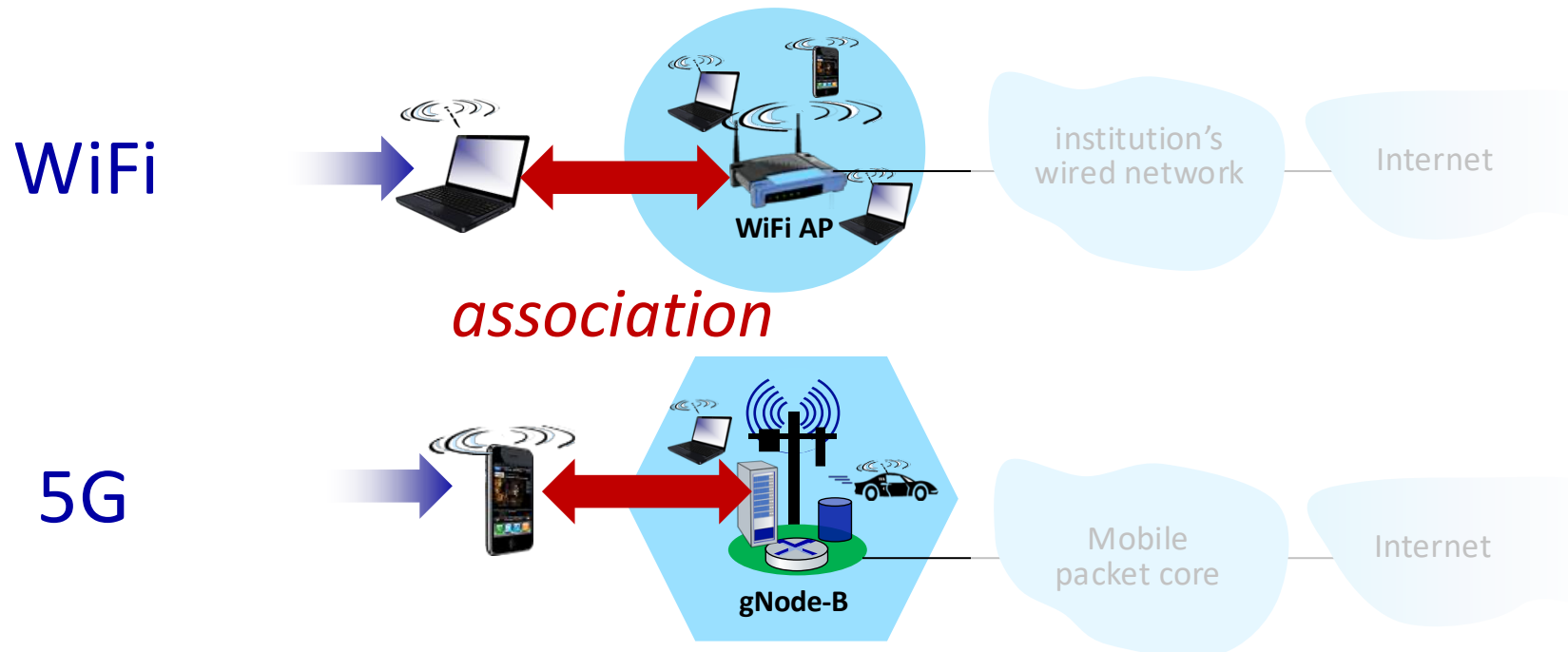
5G Core Network

- Core situated between RAN and other endpoints (Core, larger Internet)
 - single Core; multiple RANs
- consist of links, routers, servers, providing services to devices and base stations
 - “all IP” Core, but very different services than traditional Internet apps
- clear logical separation between control-plane, user plane:
 - **CUPS**: Control-Plane and User-Plane Separation



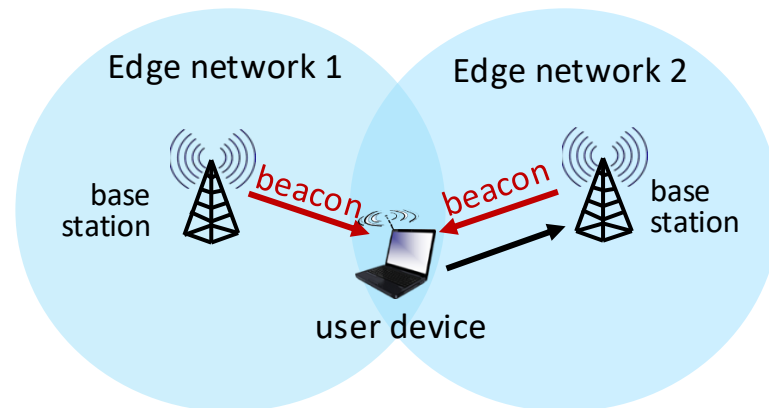
Joining the edge wireless network

- *arriving device*: must establish initial connection with node (WiFi AP, 4G/5G base station) at *wireless edge* of network
 - initial connection then used for authentication, DHCP, gain full network access
 - “bootstrapping” connectivity: *initial connection is just with edge node*

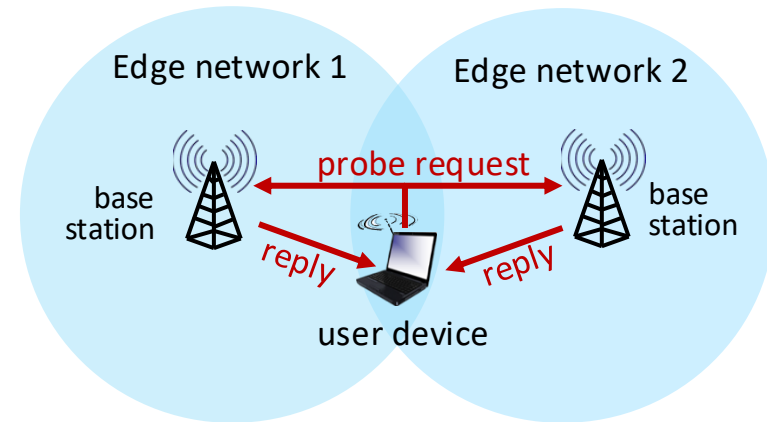


Association: two approaches

Beaconing: mobile listens to wireless channels for “beacons” advertising existence of wireless network node(s)



Probing: mobile actively transmits probe messages, waits for wireless network node(s) to reply



Either (or both) approaches used in practice in an edge network

What is mobility?

- spectrum of mobility, from the **network** perspective:

no mobility

high mobility



device moves between networks, but powers down while moving

device moves within same WLAN/RAN in one provider network

device moves among WLANs/RANs in one provider network

device moves among multiple provider networks, while maintaining ongoing connections

We're interested in these!

Bluetooth (BT) overview

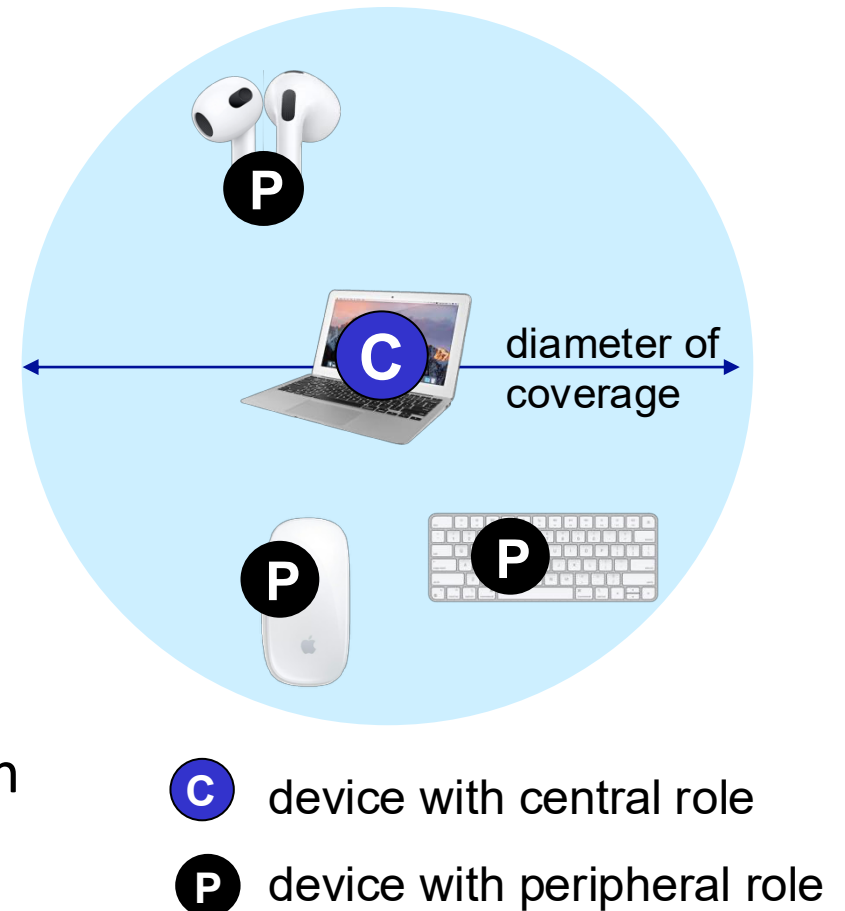
example of *wireless ad hoc network*: devices have no “infrastructure” (e.g., access point, base station) to connect to

- BT devices must find other BT devices, *organize themselves* into a network from scratch!
 - connect devices at link layer, providing channel allocation
 - addressing assignment
 - service discovery
 - application-to-application communication over Bluetooth network
 - security

*An entirely **new** architecture and protocol stack (different from Internet) is needed!*

Bluetooth: overview

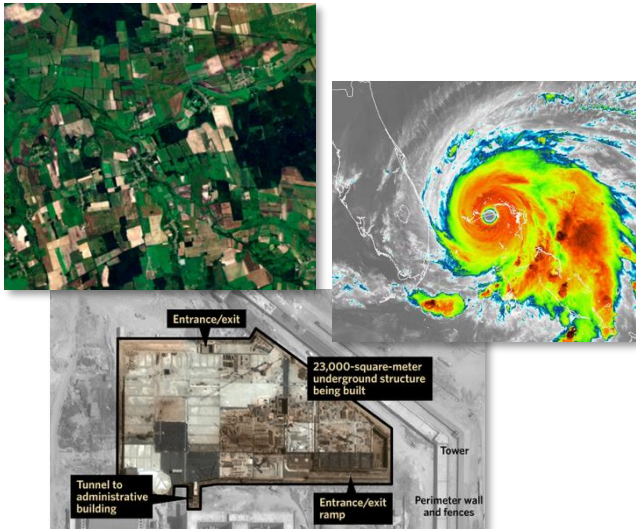
- Bluetooth network: **piconet**
- coverage: generally <10 m diameter
- no more than 8 devices per piconet
 - device initially forming network: **central** role
 - up to 7 more devices: **peripheral** role
- *communication only between central and peripheral node*
 - no direct peripheral-to-peripheral communication
- BT device has unique 48-bit address
- Operate in unlicensed ISM band: 2.4 GHz



Satellite applications

Sensing:

- environment: weather, land use, various human activity
- resolution: ~m/pixel



Broadcast:

- consumer: Direct to Home TV/video, radio (Dish, SiriusXM)
- business: content to many cable head ends
- GPS
- leverage broadcast: one send reaches many users

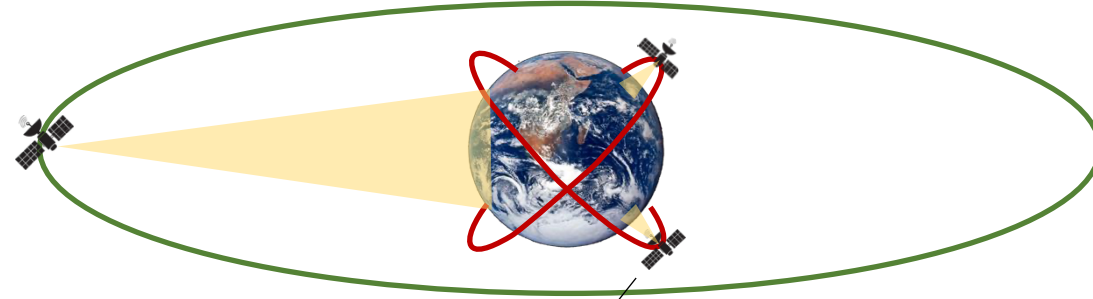


Internet:

- connectivity to unserved regions, alternative to wired Internet
- low latency over long distances
- access in airplanes



GEO and LEO satellites

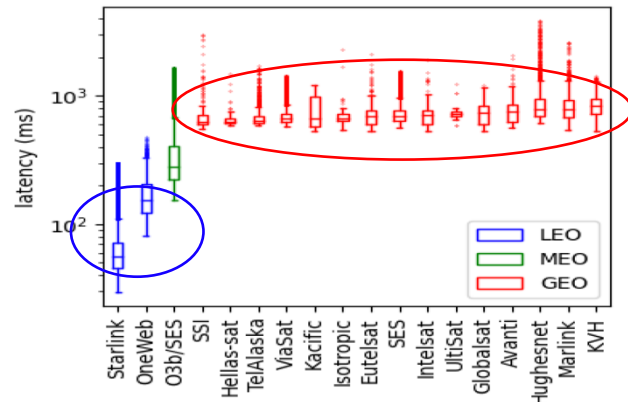


GEO:

- *geosynchronous*: stationary with respect to ground
- 35,768 km above earth
- ~800 msec RTT
- wide area coverage / satellite

LEO:

- *Low Earth Orbit*: satellite moves with respect to ground: 27,000 km/hour
 - within line of sight (LOS) of ground station for 5-15 mins
- 550 - 1200 km above earth
- ~30 msec RTT in practice
- smaller area coverage / satellite
 - constellation (network) of multiple LEO satellites to cover wide area
 - inter-satellite links (ISL)



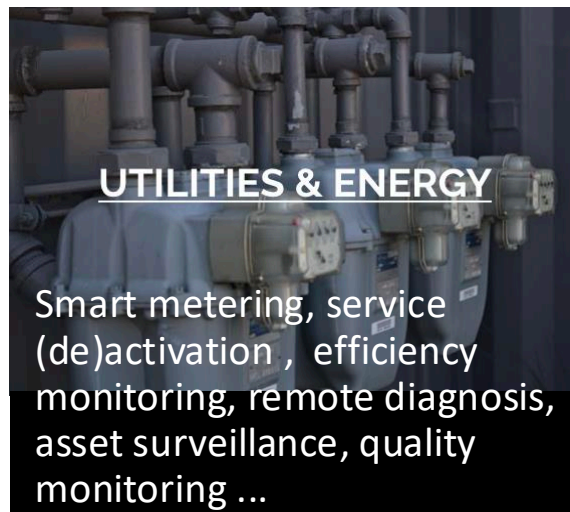
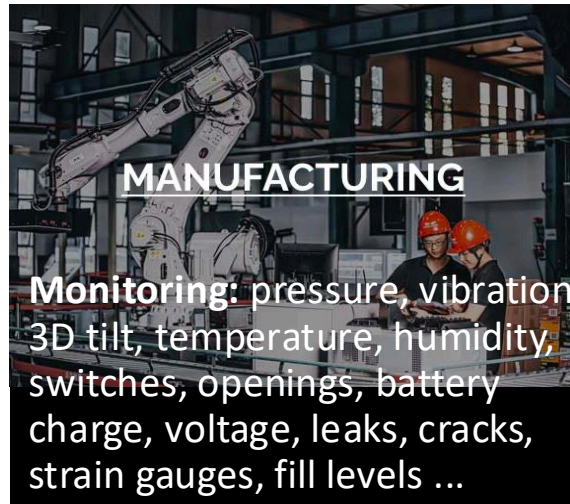
Starlink

- April 2025: ~7,239 operational LEO satellites (planned: 12K, requested: 30K)
 - first launches: 2019
 - three low-Earth-orbit [orbital shells](#), at 525, 530, 535 km
 - satellites: 500 – 2700 lbs
- services:
 - Internet service: 4M subscribers in Sept 2024 (residential US: \$120/mo)
 - service in Ukraine (terrestrial networks damaged)
 - 2024: testing direct-to-smartphone tests would use cellular spectrum from SpaceX's U.S. mobile partner T-Mobile



60 Starlink satellites stacked together before deployment on 24 May 2019

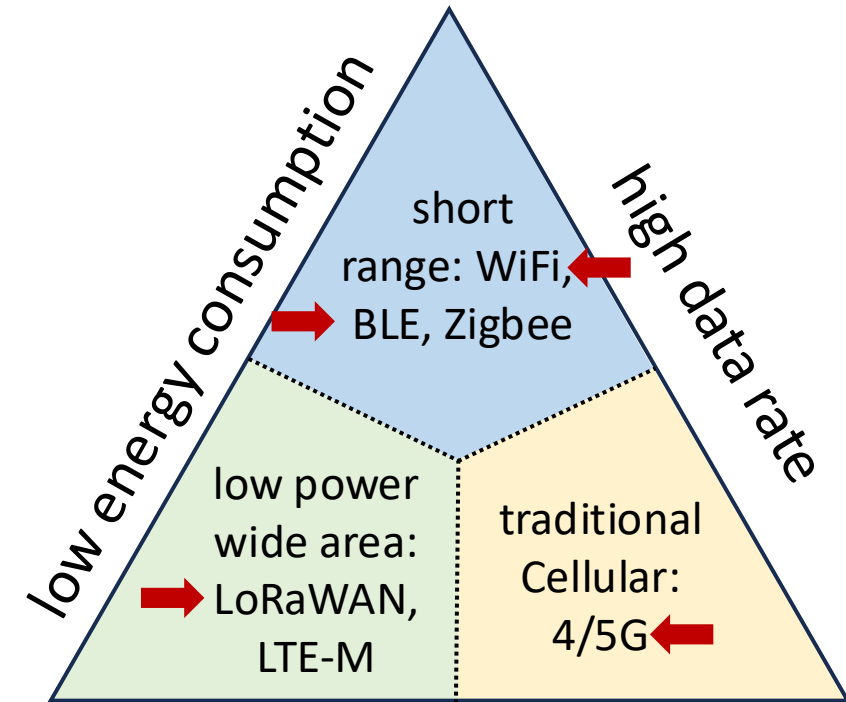
IoT: many, many use cases!



.... and there's many more!

IoT communication requirements

- **wireless**
- **distance:** from meters to kilometers
- **data rate:** 100's bps when active (metering) to ~1 Mbps (video surveillance)
- **energy:** always a consideration
 - up ~10 years without external power
 - local energy harvesting



IoT technical tradeoffs